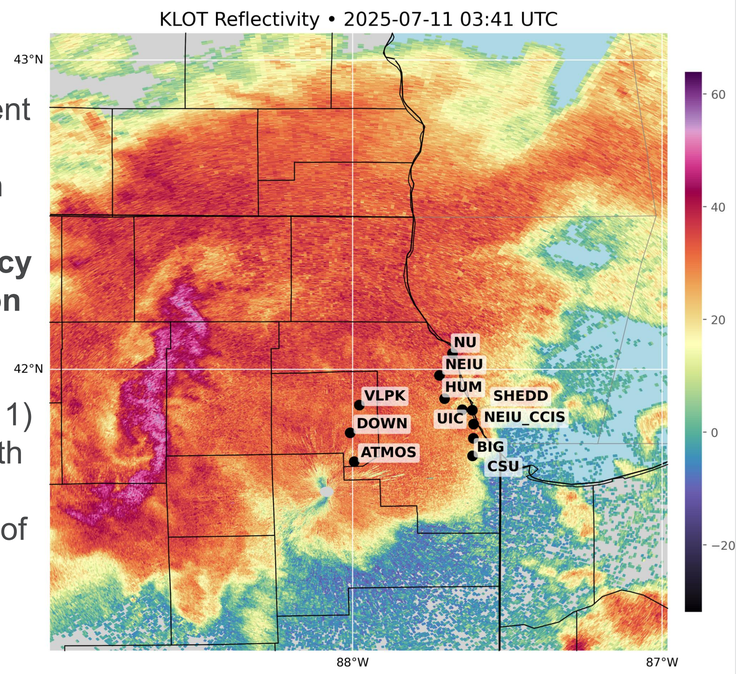
PREDICTION OF RADAR-DERIVED QUANTITATIVE PRECIPITATION ESTIMATION USING BASIC MACHINE LEARNING TECHNIQUES

Machine learning to correct radar-derived precipitation estimates.

Brandon Weart, Robert Jackson, Environmental Science Division, Argonne National Laboratory

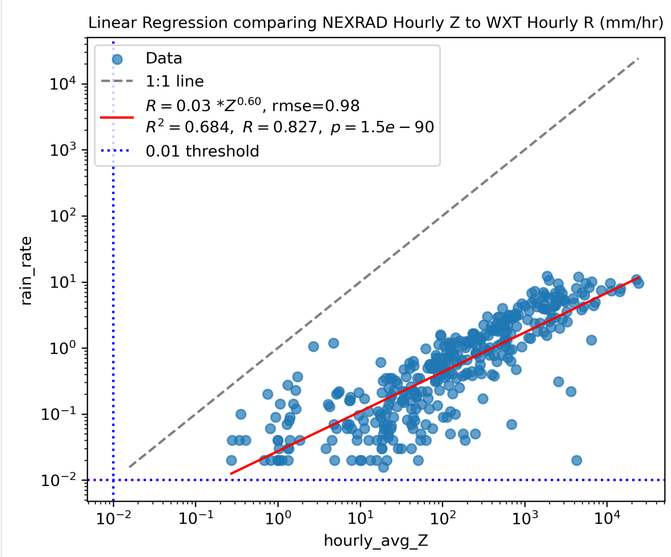
MOTIVATION

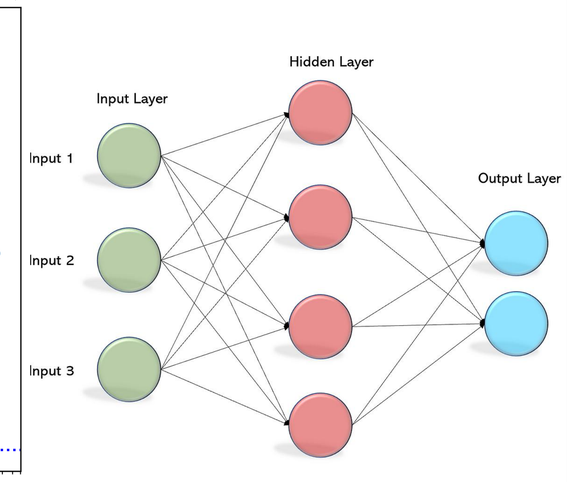
- Radar-derived precipitation estimates are prone to large inaccuracies due to the inherent biases produced by the radar. The question here is **how can basic machine learning techniques increase accuracy of radar-derived precipitation estimation.**
- CROCUS Micronet sites (Fig. 1) at ATMOS and NEIU along with the WSR-88D at KLOT were used, and days from summer of 2024, and the duration of the CROCUS Urban Rainfall campaign (March-Early June 2025) were utilized.



METHODS

- Linear regression used for Z-R relationship.
- Multi-layer perceptron-type neural network for adaptive relationship.
- 5-Fold CV was used to ensure model generalizability.





BRIEF MODEL DESCRIPTION

- Linear regression model in log-log space to account for reflectivity, which is a logarithmic variable.
- MLP model was constructed after hyperparameter tuning, which included a one-layer deep, eight-node, L2 regularized, MLP model tailored to the dataset.

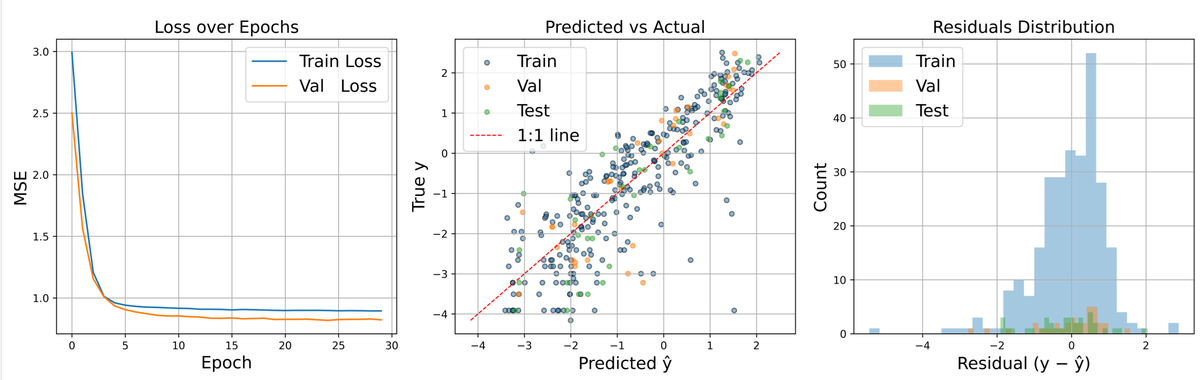
```
powers_of_two = [2**n for n in range(7)]

param_grid = {
    "mlp__n_hidden": [1, 2],
    "mlp__units": powers_of_two,
    "mlp__activation": ["relu"],
    "mlp__lr": [1e-3, 1e-4],
    "mlp__l2_lambda": [0.0, 1e-2],
    "mlp__dropout": [0.0, 0.2],
    "mlp__batch_size": powers_of_two,
    "mlp__epochs": [50, 100],
}
```

```
model = models.Sequential(
    [
        layers.Input(shape=(X_tr_scaled.shape[1],)),
        layers.Dense(8, activation="relu", kernel_regularizer=regularizers.l2(0.01)),
        layers.Dense(1),
    ]
)
model.compile(optimizer=Adam(learning_rate=1e-3), loss="mean_squared_error")
```

RESULTS

- Linear regression Z-R showed notable improvements in convective storms, but not for stratiform events.
- Using the empirical Z-R relationship, the multi-layer-perceptron (MLP) predicted R from Z with similar accuracy to linear regression (Fig. 3).
- Applying the linear regression model to in-situ cases during the week of July 7th-12th (Fig. 4) also resulted in notable improvements compared to the empirical Z-R relationship.
- Dataset imbalance led to decreased ability for the MLP to generalize, leading to decreasing performance from validation to testing. On an event basis, the performance of both models increase, especially the MLP.



Methods	RMSE mm/hr	R^2	R
Original Z-R	1.411	0.635	0.812
Lin. Reg. Z-R	1.64	.503	.845
MLP Predictor (Test split)	.9094	.7544	.8686

CONCLUSIONS

- The MLP model with its current configuration is well-equipped to predict and correct radar-derived rainfall rates, even over a large timespan.
- The linear regression Z-R method performs slightly worse overall compared to the empirical Z-R relationship at correcting over a larger timespan, but as mentioned (not shown), its performance is noticeably better on an individual event basis.

NEXT STEPS

- Implementation of the linear regression method into the existing network of CROCUS Micronet nodes for more accurate depictions of radar-derived rainfall for the location and surrounding areas, as well as feeding that data to the NWS to decrease the time needed to issue warnings for flooding events to avoid catastrophes such as the recent event in Texas.
- Develop a much more accurate prediction model with more advanced learning techniques and possibly comparing to the MRMS derived rainfall

REFERENCES

- J. Huangfu, Z. Hu, J. Zheng, L. Wang, and Y. Zhu, "Study on Quantitative precipitation estimation by Polarimetric Radar using Deep Learning," Advances in Atmospheric Sciences **41**(6), 1147–1160 (2024).
- T. Yo, S. Su, J. Chu, C. Chang, and H. Kuo, "A deep learning approach to Radar-Based QPE," Earth and Space Science **8**(3), (2021).
- M. Liu, J. Zuo, J. Tan, and D. Liu, "Comparing and optimizing four machine learning approaches to Radar-Based quantitative precipitation estimation," Remote Sensing **16**(24), 4713 (2024).
- Polarimetric radar quantitative precipitation estimation using deep convolutional neural networks," IEEE Journals & Magazine | IEEE Xplore, (2023).

U.S. DEPARTMENT of ENERGY

Argonne National Laboratory is a U.S. Department of Energy laboratory managed by UChicago Argonne, LLC.

This material is based upon work supported by the U.S. Department of Energy, Office of Science, Office of Biological and Environmental Research's Urban Integrated Field Laboratories CROCUS project research activity, under Contract Number DE-AC02- 06CH11357 .

Argonne NATIONAL LABORATORY

Abstract

The advent of artificial intelligence and machine learning within the atmospheric science community has allowed for the continued expansion of boundaries established by the current state of computational power. From emulating climate models with half the normal computing power to detecting features on radar, machine learning has become one of the most essential tools for research in atmospheric science. This study utilizes machine learning to predict precipitation derived from the KLOT NEXRAD radar, which covers the Chicago area. Radar-derived rainfall is known to have issues with accuracy, as radar tends to be biased towards larger drop sizes, making estimations shaky at best for lower-intensity stratiform events with smaller drop sizes. Machine learning algorithms were applied to radar-derived and ground-truth precipitation data, including basic and multiple linear regression in log-log space, as well as a multi-layer perceptron neural network. These algorithms were developed and evaluated using standard statistical performance metrics. Success was achieved when applying the multilayer perceptron on a dataset of ~400 points, with an RMSE of .9303 and correlation (R) of .86. Applying machine learning models to simple tasks such as precipitation estimation allows for the retrieval of patterns or correlations in data that may be difficult to spot with more conventional methods.